Christopher Yu

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Skills

Active Clearances: DoD Final Top Secret
Languages: Python, C++/C, JavaScript, SQL
Technologies/Platforms: Kubernetes, Docker, Redis, MySQL, Git, Bash, Linux
Frameworks/Tools: Pytorch, NumPy, FastAPI, Temporal, Pytest, Postman, Pandas, React, Minikube

Experience

Research Engineer**, Johns Hopkins University Applied Physics Laboratory –** Laurel, Maryland July 2025 – Present

Subliminal Learning Research

- Investigated implicit behavioral trait transfer via model distillation across natural language and source code
- Locate and apply steering vectors per different model/transformer architectures to induce model biases and generate training data for gradient-based distillation into student models
- Measure bias transfer across model/transformer architectures, comparing steering induced transfer against system prompt conditioning

Internal Offensive AI Research

- Led red-team assessment of vLLM inference platform integrity and data exfiltration risk from adversaries with telemetry endpoint access in tool-enabled agentic workflows (eg. Claude Code)
- Developed reusable attack prototypes to support adversarial testing of inference-stack integrity and evaluate defensive detection and mitigation strategies
- Generated reproducible range with multiple gateways and harnesses demonstrating vulnerable transition to hardware

Agentic Mission Planner

- Designed verification framework for agentic mission planners coordinating heterogeneous UxS and EW assets across internal and industry autonomy systems
- Developed agent evaluations and observability pipelines using Inspect AI, Arize Phoenix and Temporal to visualize agent trajectory and validate MILP solver outputs with LLM-as-judge to guarantee defensible planning decisions
- Investigated DSPy GEPA for evaluation driven prompt optimization to improve planning optimality, feasibility and alignment with operational constraints

SWE Intern**, Johns Hopkins University Applied Physics Laboratory** – Laurel, Maryland May 2024 – Jul 2025

- Designed multi-agent adversarial simulation environment in PettingZoo to analyze resilience of systems under attack
- Implemented MiniMax Q-Learning in Python and conducted hyperparameter tuning to identify common attack types on systems (eg. supply chain attacks), identify valuable assets and uncover critical failure points on SysML models

SWE Intern**, NYU Langone Health** – New York, NY June 2022 – August 2022

- Automated Human Leukocyte Antigen (HLA) DNA analysis in pre-production cloud-based web app via translating physician workflows into optimized algorithms and coordinating development through JIRA and biweekly standups
- Formulated frontend components via React and implemented backend models with API endpoints for compatibility algorithm, utilizing MySQL queries in the backend to draw various proteins and data from the database

Projects

Aerial Zero Trust - Master's Thesis https://github.com/cyu72/AZT

Christopher Yu, Ping Yang, Jayson Boubin, "Aerial Zero Trust", Dronesys, 2026

- Conceptualized and implemented protocol designed to secure drone swarm communications through authentication, verification and obfuscation based on prior work in Mobile Ad-Hoc Networks (MANETs) for a co-authored paper published at IEEE-SEC
- Developed a Kubernetes-based simulation using Minikube, C++, and Python, leveraging Container Network Interfaces (CNIs) and services to replicate real-world drone topologies, enabling scalable testing of networks with up to 50 nodes

Education

SUNY Binghamton – MS in Computer Science & Cybersecurity May 2025
SUNY Binghamton – BS in Computer Science May 2024